THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 5 · Term 3

Student edition

Outcomes: MA-5.6 distance, time and the unitary method; MA-5.7 angles and triangles; MA-5.8 area and perimeter; MA-5.9 averages and bar graphs.

Lesson notes

MA-5.6 Distance, time and unitary method (Weeks 1–4)

Speed = distance ÷ time; find distance or time from the other two (bus journeys between Pakistani cities). Unitary method: find the value of one, then of many (5 pens cost Rs 150, so 1 costs Rs 30 and 8 cost Rs 240).

MA-5.7 Angles and triangles (Weeks 5–8)

Straight angle = 180°; reflex angle between 180° and 360°; a full turn = 360°. Classify triangles by sides (equilateral, isosceles, scalene) and by angles (acute, right, obtuse). Tear the corners of a paper triangle to show the angles make 180°.

MA-5.8 and MA-5.9 Area, perimeter, data (Weeks 9–10)

Area of a rectangle = length × breadth; square = side × side; perimeter formulas; areas of shapes made of rectangles. Average = total ÷ number of values; read horizontal and vertical bar graphs with scales.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Angles in a triangle** — Change the shape of the triangle. The three angles always add up to the same total.

Workbook pages

Page	Outcome	Activity
1–2	MA-5.6	Journey problems; unitary-method shopping.
3–4	MA-5.7	Classify triangles; find missing angles.
5–6	MA-5.8	Area and perimeter of the classroom and an L-shaped garden.
7	MA-5.9	Class test-score averages; bar graph with a scale of 5.

Question bank

#	Outcome	Question	Your answer
1	MA-5.6	A bus travels 120 km in 2 hours. Its speed is …	240 km/h / 60 km/h / 122 km/h / 30 km/h

#	Outcome	Question	Your answer
2	MA-5.6	At 60 km/h, how long to travel 180 km?	2 h / 3 h / 4 h / 120 h
3	MA-5.6	5 pens cost Rs 150. What do 8 pens cost?	Rs 200 / Rs 240 / Rs 300 / Rs 1,200
4	MA-5.7	A straight angle is …	90° / 180° / 270° / 360°
5	MA-5.7	An angle of 250° is …	obtuse / straight / reflex / acute
6	MA-5.7	A triangle has angles 50° and 60°. The third angle is …	60° / 70° / 80° / 110°
7	MA-5.7	A triangle with all sides equal is …	isosceles / scalene / equilateral / right-angled
8	MA-5.7	A triangle can have two right angles. True or false.	True / False
9	MA-5.8	Area of a rectangle 8 cm by 5 cm =	13 cm² / 26 cm² / 40 cm² / 40 cm
10	MA-5.8	Perimeter of the same rectangle =	13 cm / 26 cm / 40 cm / 80 cm
11	MA-5.8	Area of a square with 6 cm sides =	24 cm² / 36 cm² / 12 cm² / 36 cm
12	MA-5.9	Average of 4, 7, 9, 12, 8 =	7 / 8 / 9 / 40
13	MA-5.9	In a bar graph with a scale of 5, a bar reaching the 4th line shows …	4 / 9 / 20 / 45